

Time & Work Pipes & Cisterns LCM, HCF Approach - 01

A can do a piece of work in 12 days whereas B can do the same work in 18 days. In how many days (A & B) together can do the same work if they are working together?

Sol: LCM of 12 & 18 $\rightarrow 36$
 efficiency, $N_A = \frac{36}{12} = 3$
 $N_B = \frac{36}{18} = 2$
 $(A+B) = \frac{36}{3+2} = 7\frac{1}{5}$ days ✓

Total no. of Prime Numbers: 1 to 60 → prime no.
 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 59
 51 to 100 → 53, 59, 61, 67, 71, 73, 79, 83, 89, 97, 101

LCM = Least Common Multiple
 24 $\rightarrow 2^3 \times 3^1$
 45 $\rightarrow 3^2 \times 5^1$
 LCM $\rightarrow 2^3 \times 3^2 \times 5^1 = 360$
 HCF $\rightarrow$ select least power of prime base
 $HCF \rightarrow 3^1 \times 5^1 = 15$

Q) A can do one fifth part of a job in 4 days, B can do one sixth part of the same work in 5 days. If they work together, then in how many days can the job be completed?

A $\rightarrow \frac{1}{5}$ in part $\rightarrow 4$ days
 B $\rightarrow \frac{1}{6}$ in part $\rightarrow 5$ days
 $A \rightarrow 5 \times 4 = 20$ days
 $B \rightarrow 5 \times 6 = 30$ days
 LCM(20, 30) $\rightarrow 60$
 $N_A = \frac{60}{20} = 3$
 $N_B = \frac{60}{30} = 2$
 $(A+B) = \frac{60}{3+2} = 12$ days

Q) A alone can do work in 30 days. B alone can do the same work in 60 days. If they work together for 5 days, then what part of the work will be left?

A $\rightarrow 30$ days
 B $\rightarrow 60$ days
 LCM(30, 60) $\rightarrow 60$
 $N_A = \frac{60}{30} = 2$
 $N_B = \frac{60}{60} = 1$
 $(A+B) = \frac{60}{2+1} = 20$ days
 In 5 days (A+B) complete how many units?
 $\Rightarrow (N_A + N_B) \times 5 = (2+1) \times 5 = 15$ units
 Remaining units $\Rightarrow 60 - 15 = 45$ units

Q) LCM of 48, 50, 98, 54 and 72
 $\rightarrow$ solving speed will depend on options also (by eliminating)

How to find LCM & HCF of fractions
 $LCM\left[\frac{a}{b}, \frac{c}{d}, \frac{e}{f}\right] \Rightarrow \frac{LCM[a, c, e]}{HCF[b, d, f]}$
 $HCF\left[\frac{a}{b}, \frac{c}{d}, \frac{e}{f}\right] \Rightarrow \frac{HCF[a, c, e]}{LCM[b, d, f]}$

Q) What is the LCM of 3.6, 1.8 & 0.144
 Step 1 $\rightarrow$ try to make denominator same
 Step 2 $\rightarrow$ then use $LCM\left[\frac{a}{b}, \frac{c}{d}, \frac{e}{f}\right]$ formula

$\hookrightarrow$ solve LCM, HCF may give to get faster solution

Q) Two numbers are in the ratio 7:11. If their HCF is 28, then the sum of two numbers is:
 A $\rightarrow 20$
 B $\rightarrow 25$
 C $\rightarrow 40$
 D $\rightarrow 45$
 Sol: 7:11 ratio, HCF = 28
 Numbers are $7 \times 28 = 196$ and $11 \times 28 = 308$
 Sum = $196 + 308 = 504$

Back to Que:
 $N_1 : N_2 = 7 : 11$
 $7 \times 18 : 11 \times 28 = 126 : 308$
 $\Rightarrow$ Sum of $N_1, N_2 = 28 \times 18 = 504$
 The traffic light at 3 different road crossings changes after every 48 sec, 72 sec, 108 sec respectively. If they all change simultaneously at 8:20 am, then at what time will they next change together simultaneously?
 In this type of question just find the LCM of [48, 72, 108]
 $48 \rightarrow 2^4 \times 3$
 $72 \rightarrow 2^3 \times 3^2$
 $108 \rightarrow 2^2 \times 3^3$
 LCM $\Rightarrow 2^4 \times 3^3 = 432$
 after 432 sec it will blink simultaneously
 7 min 12 sec

Q) If A can do a piece of work in 20 days whereas (A & B) together can be the same piece of work in 15 days. In how many days B alone can do the same work?

A $\rightarrow 20$ days
 A+B $\rightarrow 15$ days
 $N_A = \frac{60}{20} = 3$
 $N_{A+B} = \frac{60}{15} = 4$
 $N_B = 4 - 3 = 1$
 B-time = $\frac{60}{1} = 60$ days

Q) A and B can complete a work in 15 days and 10 days respectively. They started doing the work together but after 4 days B had to leave. Then A working with a new worker C completed the remaining work in 3 days. If C works alone, in how many days he can do 40% of the same work?

Part 1: find the N of A, B, C
 A $\rightarrow 15$ days $\rightarrow 4$ units
 B $\rightarrow 10$ days $\rightarrow 6$ units
 LCM $\rightarrow 30$
 $N_A = \frac{30}{15} = 2$
 $N_B = \frac{30}{10} = 3$
 Part 2: $(N_A + N_B) \times 4 = 20$ units
 Remaining = $30 - 20 = 10$ units
 $(N_A + N_C) \times 3 = 10$
 $(2 + N_C) \times 3 = 10$
 $N_C = \frac{10}{3} - 2 = \frac{4}{3}$
 $N_C = \frac{4}{3}$ units/day
 40% of total work $\Rightarrow \frac{40}{100} \times 30 = 12$ units
 Time = $\frac{\text{work}}{\text{efficiency}} = \frac{\text{units}}{\text{units/day}} = \frac{12}{4/3} = 9$ days

A can complete a work in 24 days, B can complete the same work in 32 days. If they work together and after certain time B left, then work and A completed the remaining work. Then total work be completed in 18 days. After how many days B left the work?

A $\rightarrow 24$ days $\rightarrow 12$ units
 B $\rightarrow 32$ days $\rightarrow 9$ units
 LCM $\rightarrow 96$
 $N_A = 4$
 $N_B = 3$
 A $\rightarrow 18$ days $\Rightarrow N_A \times 18 = 4 \times 18 = 72$ units

Part to total work = 96
 $B(\text{work}) = 96 - 72 = 24$ units
 $B(\text{time}) = \frac{24 \text{ units}}{3} = 8$ days

Q) A tap can fill a cistern in 8 hrs and another can empty in 16 hrs. If both the taps are opened simultaneously the time (in hrs) to fill the cistern will be
 A $\rightarrow 8$
 B $\rightarrow 16$
 C $\rightarrow 10$
 D $\rightarrow 12$
 Sol: $N_A = 12$ units/hr
 $N_B = -6$ units/hr
 $(A+B) = \frac{96}{12-6} = 16$ hrs

Q) A cistern has two taps which fill it in 12 min & 15 min respectively. There is also a waste pipe in the cistern. When all pipes are opened, the empty cistern is full in 20 min. How long will the waste pipe take to empty a full cistern?

A $\rightarrow 12$ min
 B $\rightarrow 15$ min
 C $\rightarrow 20$ min
 D $\rightarrow 30$ min
 Sol: $N_A = 8$ units/min
 $N_B = 4$ units/min
 $N_{A+B+C} = \frac{60}{20} = 3$
 $N_C = 3 - 8 - 4 = -9$
 Time = $\frac{60}{9} = 6\frac{2}{3}$ min

Q) Pipe A can fill a tank in 12 hrs whereas Pipe B can empty the same tank in 20 hrs. If both the pipes are operating simultaneously, then in how many hrs half of the tank gets filled?

A $\rightarrow 12$ hrs
 B $\rightarrow 20$ hrs
 Sol: $N_A = 5$ units/hr
 $N_B = -3$ units/hr
 $(A+B) = \frac{60}{5-3} = 30$ hrs
 Time = $\frac{60}{2} = 30$ hrs for full tank
 $\Rightarrow 15$ hrs for half